Listing 3.3 A minimal HTML page that uses the client side Javascript AWS SDK in an HTML page to publish an MQTT message containing measured swimming pool pH.

```
<html>
<body>
  <input type="number" id="phValue" placeholder="pH value">
  <button onclick="publish()">Send</button>

  <script src="https://cdnjs.cloudflare.com/ajax/
    libs/aws-sdk/2.1400.0/aws-sdk.min.js"></script> #1
  <script>
    AWS.config.update({ #2
      region: REGION,
      credentials: new AWS.CognitoIdentityCredentials({
        IdentityPoolId: IDENTITY_POOL_ID
      })
    });

    const iot = new AWS.IotData(
      {endpoint: MQTT_ENDPOINT}); #3

    function publish() {
      iot.publish({ #4
        topic: 'home/water/pool/chemistry_sensor_1',
        payload: JSON.stringify({ph: phValue.value})
      }, (e,d) => console.log(e||'Sent'));
    }
  </script>
</body>
</html>
```

#1 Import the AWS SDK for Javascript.
#2 Configure the AWS SDK with your region, and the identifier of your Cognito identity pool.
#3 Instantiate the IoT data client.
#4 Publish the value entered into the input field to the pool water chemistry MQTT topic.

MAKING MANUAL COLLECTION SUSTAINABLE

The biggest challenge with manual data collection is the human factor of ensuring consistent and accurate collection. If the process is difficult or tedious, or the value of the collecting the data is not apparent, it is likely to be abandoned. To create a sustainable manual data collection process, keep the following in mind: